

Game Theory, Mres

Homework

Jordi Torres-Vallverdú

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1.1

Proof. Suppose it is. Then by the definition of properness there is a sequence of totally mixed profiles $\sigma^n \rightarrow (A, V)$ and reals $\varepsilon_n \rightarrow 0$, $\varepsilon_n > 0$, such that for each player i and every pair of pure strategies s, s' of that player,

$$u_i(s, \sigma_{-i}^n) < u_i(s', \sigma_{-i}^n) \implies \sigma_i^n(s) \leq \varepsilon_n \sigma_i^n(s'). \quad (*)$$

I write player 1's tremble weights as $(a_n, b_n, c_n, d_n) = (\sigma_1^n(A), \sigma_1^n(B), \sigma_1^n(C), \sigma_1^n(D))$ and player 2's as $(v_n, w_n, x_n, y_n, z_n) = (\sigma_2^n(V), \dots, \sigma_2^n(Z))$. Being totally mixed, all of these are strictly positive; and $\sigma^n \rightarrow (A, V)$ means $a_n, v_n \rightarrow 1$ while the other six weights $\rightarrow 0$.

The idea is to use the two player's constraints against each other. I do it in three steps.

1. Against player 1's trembles, player 2's payoffs from W , X and Y are

$$u_2(W, \sigma_1^n) = a_n + d_n, \quad u_2(X, \sigma_1^n) = a_n + c_n, \quad u_2(Y, \sigma_1^n) = 0,$$

because column Y gives player 2 zero in every row, while W and X each give him a 1 in row A . Since $a_n \rightarrow 1$, for n large both $a_n + d_n$ and $a_n + c_n$ are strictly positive, so Y is strictly worse than W and than X . By $(*)$,

$$y_n \leq \varepsilon_n w_n \quad \text{and} \quad y_n \leq \varepsilon_n x_n.$$

In particular $y_n < w_n$ and $y_n < x_n$ for n large (eventually $\varepsilon_n < 1$).

2. For player 1, against player 2's trembles¹:

$$u_1(B) = v + 2w + 2x + z, \quad u_1(C) = v + w + 2x + y + z, \quad u_1(D) = v + 2w + x + y + z,$$

so, cancelling, the only columns that survive are W, X, Y and

$$u_1(B) - u_1(C) = w_n - y_n, \quad u_1(B) - u_1(D) = x_n - y_n.$$

¹I suppress the superscript n for readability

By Step 1 both differences are strictly positive for n large, so C and D are each strictly worse than B . Applying $(*)$ to player 1,

$$c_n \leq \varepsilon_n b_n, \quad d_n \leq \varepsilon_n b_n, \quad \text{hence} \quad c_n + d_n \leq 2\varepsilon_n b_n < b_n$$

once $\varepsilon_n < \frac{1}{2}$.

3 Finally, step 3:

Now we compare columns V and Z for player 2:

$$u_2(V, \sigma_1^n) = a_n + b_n + c_n + d_n = 1, \quad u_2(Z, \sigma_1^n) = a_n + 2b_n,$$

so

$$u_2(V, \sigma_1^n) - u_2(Z, \sigma_1^n) = (c_n + d_n) - b_n < 0$$

for n large, by 2. Thus V is strictly worse than Z , and $(*)$ gives

$$v_n \leq \varepsilon_n z_n \leq \varepsilon_n \rightarrow 0.$$

But $\sigma^n \rightarrow (A, V)$ forces $v_n \rightarrow 1$. Contradiction. $\blacksquare$

1.2

Basically I need to show that is Sequential Equilibria, but making sure the proof or the argument is applicable to any extensive form game that we may draw.

Suppose player 2 plays V , so by (a) player 1 gets 1 no matter which row she ends up playing. Hence at every information set of player 1, every continuation — A included — yields 1, so A is (weakly) optimal there under whatever beliefs she holds. Nothing about Γ entered this; player 1's sequential rationality is automatic in any tree.

By the first half of (b), V yields a constant 1 at any information set, whatever the belief μ . So the only way V could fail to be optimal is if some other column beats 1. Writing the belief's marginal over player 1's row as $(\mu_A, \mu_B, \mu_C, \mu_D)$,

$$u_2(V) = 1, \quad u_2(W) = \mu_A + \mu_D \leq 1, \quad u_2(X) = \mu_A + \mu_C \leq 1, \quad u_2(Y) = 0, \quad u_2(Z) = \mu_A + 2\mu_B.$$

Only Z can exceed 1, and $u_2(Z) = \mu_A + 2\mu_B > 1$ exactly when $\mu_B > \mu_C + \mu_D$. So V is a best reply at an information set as soon as

$$\mu_B \leq \mu_C + \mu_D,$$

i.e. as long as the belief doesn't put more weight on the B -node than on C and D together.

On the path, player 1 plays A with probability 1, so Bayes forces $\mu_A = 1$ (hence $\mu_B = 0$) at every on-path information set of player 2 — and there V is optimal, tying Z at 1. Off the path we have freedom, subject to μ being a limit of Bayes-updates from totally mixed trembles. Take any sequence $\sigma^n \rightarrow (A, V)$ of totally mixed profiles in which player 1's tremble onto B is of higher order than his trembles onto C and D — say $\sigma_1^n(B) = o(\sigma_1^n(C))$ and $\sigma_1^n(B) = o(\sigma_1^n(D))$. Then at every information set the conditional weight on the B -node vanishes, so in the limit $\mu_B = 0 \leq \mu_C + \mu_D$, and V is optimal everywhere by the line above. This recipe never mentioned the shape of Γ , so it produces a consistent, sequentially rational (σ, μ) in every tree.

Conclusion. (σ, μ) is a sequential equilibrium of Γ with $\sigma = (A, V)$, and Γ was an arbitrary extensive form with this normal form. So (A, V) is part of a sequential equilibrium in every such extensive form.

Putting the two halves together: (A, V) lies in the intersection of the sequential-equilibrium profiles taken over all extensive forms, yet (Part 1) it is not proper. Hence

$$\text{proper} \subsetneq \bigcap_{\Gamma} \text{sequential eq. of } \Gamma,$$

which is the strict inclusion the notes were pointing at.

I think intuition is correct, but proof needs to be finished and more formal.

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